



Pelvic Inflammatory Disease (PID)

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Pelvic inflammatory disease (PID) is a polymicrobial infection of the upper female genital tract: the cervix, uterus, fallopian tubes, and ovaries; abscess may occur. PID may be caused by sexually transmitted infections. Common symptoms and signs include lower abdominal pain, irregular vaginal bleeding, cervical discharge, and cervical motion tenderness. Long-term complications may include infertility, chronic pelvic pain, and ectopic pregnancy. Diagnosis includes polymerase chain reaction testing of cervical specimens for *Neisseria gonorrhoeae* and *Chlamydia trachomatis*, microscopic examination of cervical discharge (usually), and ultrasonography or laparoscopy (occasionally). Treatment is with antibiotics.

Pelvic inflammatory disease (PID) may involve the cervix, uterus, fallopian tubes, and/or ovaries. Infection of the cervix ([cervicitis](#)) causes mucopurulent discharge. Infection of the fallopian tubes (salpingitis) and uterus (endometritis) tend to occur together. If severe, infection can spread to the ovaries (oophoritis) and then the peritoneum (peritonitis). Salpingitis with endometritis and oophoritis, with or without peritonitis, is often called salpingitis even though other structures are involved. Pus may collect in the tubes (pyosalpinx), and an abscess may form (tubo-ovarian abscess).

Etiology of PID

PID results from microorganisms ascending from the vagina and cervix into the uterus and fallopian tubes. The sexually transmitted infections [Neisseria gonorrhoeae](#) and [Chlamydia trachomatis](#) are common causes of PID. *Mycoplasma genitalium*, which is also sexually transmitted, can also cause or contribute to PID. Incidence of sexually transmitted PID is decreasing; < 50% of patients with acute PID test positive for gonorrhea or chlamydia.

PID usually also involves other aerobic and anaerobic bacteria, including pathogens that are associated with [bacterial vaginosis](#). Vaginal microorganisms such as *Haemophilus influenzae*, *Streptococcus agalactiae*, and enteric gram-negative bacilli can be involved in PID, as can *Ureaplasma* species. Vaginal inflammation and bacterial vaginosis help in the ascending spread of vaginal microorganisms.

Risk factors

Pelvic inflammatory disease commonly occurs in women < 35 years old. It is rare before menarche, after menopause, and during pregnancy.

Risk factors include

- Previous PID
- Presence of bacterial vaginosis or a sexually transmitted infection

Other risk factors, particularly for gonorrheal or chlamydial infection, include

- Younger age
- Nonwhite race
- Low socioeconomic status
- Multiple or new sex partners or a male partner who does not use a condom
- Douching

Symptoms and Signs of PID

Pelvic inflammatory disease commonly causes lower abdominal pain, fever, cervical discharge, and abnormal uterine bleeding, particularly during or after menses.

Cervicitis

In [cervicitis](#), the cervix is erythematous and friable (bleeds easily). Mucopurulent discharge is common; usually, it is yellow-green and can be seen exuding from the endocervical canal.

Acute salpingitis

Lower abdominal pain is usually present and bilateral but may be unilateral, even when both tubes are involved. Pain may also occur in the upper abdomen. Nausea and vomiting are common when pain is severe. Irregular bleeding (caused by endometritis) and/or fever each occur in up to one third of patients.

Occasionally, dyspareunia or dysuria occurs.

Even women with inflammation that is severe enough to cause scarring have minimal or no symptoms.

In the early stages, signs may be mild or absent. Later, cervical motion tenderness, guarding, and rebound tenderness are common.

PID due to *N. gonorrhoeae* is usually more acute and causes more severe symptoms than that due to *C. trachomatis*, which can be indolent. PID due to *M. genitalium*, like that due to *C. trachomatis*, is also mild and should be considered in women who do not respond to first-line therapy for PID.

Complications

Fitz-Hugh-Curtis syndrome (perihepatitis that causes upper right quadrant pain) may result from acute gonococcal or chlamydial salpingitis. Infection may become chronic, characterized by intermittent exacerbations and remissions.

A **tubo-ovarian abscess** (collection of pus in the adnexa) develops in about 15% of women with salpingitis. It can accompany acute or chronic infection and is more likely if treatment is late or incomplete. Pain, fever, and peritoneal signs are usually present and may be severe. An adnexal mass may be palpable, although extreme tenderness may limit the examination. The abscess may rupture, causing progressively severe symptoms and possibly septic shock.

Hydrosalpinx is fimbrial obstruction and tubal distention with nonpurulent fluid; it is usually asymptomatic but can cause pelvic pressure, chronic pelvic pain, dyspareunia, and/or infertility.

Salpingitis may cause **tubal scarring and adhesions**, which commonly result in chronic pelvic pain, infertility, and increased risk of ectopic pregnancy.

Diagnosis of PID

- High index of suspicion
- Pelvic examination
- Cervical tests for *N. gonorrhoeae* and *C. trachomatis*

Index of suspicion should be high, particularly in reproductive-age women and girls with risk factors, because clinical presentation is variable and even minimally symptomatic infection may have severe sequelae.

PID is suspected when women of reproductive age have lower abdominal pain or cervical or unexplained vaginal discharge, particularly with fever. PID is also considered when irregular vaginal bleeding, dyspareunia, or dysuria is unexplained.

A presumptive diagnosis of PID should be made and treatment initiated for sexually active young women and other women at risk for STIs if they are experiencing pelvic or lower abdominal pain, no cause for the illness other than PID can be identified, and if 1 or more of the following 3 minimum clinical criteria are present on pelvic examination:

- Cervical motion tenderness
- Uterine tenderness
- Adnexal tenderness

In addition, on pelvic examination, a palpable adnexal mass suggests tubo-ovarian abscess.

If PID is suspected, NAAT (nucleic acid amplification test) of cervical specimens for *N. gonorrhoeae* and *C. trachomatis* (which is about 99% sensitive and specific) is done. If NAAT is unavailable, cultures are done. However, upper tract infection is possible even if tests for cervical infection are negative.

The white blood cell count may be elevated but is not helpful diagnostically. A pregnancy test should be performed in reproductive-age women.

If a patient cannot be adequately examined because of tenderness, imaging studies are done as soon as possible. Ultrasonography is the preferred first-line test. If ultrasound is not available or is inconclusive, other imaging modalities like CT scan may be performed to evaluate for abscess formation, including tubo-ovarian abscess.

Other indications for imaging are if an adnexal or pelvic mass is suspected clinically or if patients do not respond to antibiotics within 48 to 72 hours. In such cases, ultrasonography or CT scan is done as soon as possible to exclude tubo-ovarian abscess, pyosalpinx, and disorders unrelated to PID (eg, [ectopic pregnancy](#), [adnexal torsion](#)).

If the diagnosis is uncertain after ultrasonography or other imaging modalities, or if empiric treatment for PID fails, laparoscopy should be done; purulent peritoneal material noted during laparoscopy is the diagnostic gold standard.

Pearls & Pitfalls

- If clinical findings suggest PID but the pregnancy test is positive, evaluate for ectopic pregnancy.

Differential diagnosis

If a pregnancy test is positive, [ectopic pregnancy](#), which can produce similar findings, should be considered.

Other common causes of [pelvic pain](#) include [endometriosis](#), [adnexal torsion](#), [ovarian cyst rupture](#), and [appendicitis](#). Differentiating features of these disorders are discussed elsewhere in The Manual.

Fitz-Hugh-Curtis syndrome may mimic [acute cholecystitis](#) but can usually be differentiated by evidence of salpingitis during pelvic examination or, if necessary, with ultrasonography.

Treatment of PID

- Antibiotics to cover *N. gonorrhoeae*, *C. trachomatis*, and organisms in the vaginal floras

When a PID diagnosis is suspected but cervical infection has not been confirmed or the patient does not meet all clinical criteria, empiric treatment of PID is given for several reasons:

- Test results may take a few days.
- Diagnosis based on clinical criteria can be inaccurate.
- Not treating minimally symptomatic PID can result in serious complications.

Antibiotics are given empirically to cover *N. gonorrhoeae*, *C. trachomatis*, and vaginal flora, including anaerobes, and are modified based on laboratory test results.

Pearls & Pitfalls

- Treat empirically for PID whenever the diagnosis is suspected, because test results may take time and be inconclusive, diagnosis based on clinical criteria can be inaccurate, and not treating minimally symptomatic PID can result in serious complications.

Patients with cervicitis or clinically mild to moderate PID do not require hospitalization. Outpatient treatment regimens (see table [Regimens for Treatment of Pelvic Inflammatory Disease](#)) usually also aim to eradicate [bacterial vaginosis](#), which often coexists.

Sex partners of patients with *N. gonorrhoeae* or *C. trachomatis* infection should be treated.

TABLE

Regimens for Treatment of Pelvic Inflammatory Disease*

Treatment	Recommended Regimens	Alternative Regimens
Parenteral†	Regimen A: Ceftriaxone 1 g IV every 24 hours PLUS Doxycycline 100 mg orally or IV every 12 hours PLUS Metronidazole 500 mg orally or IV every 12 hours	Regimen C: Ampicillin/sulbactam 3 g IV every 6 hours PLUS Doxycycline 100 mg orally or IV every 12 hours Regimen D: Clindamycin 900 mg IV every 8 hours PLUS Gentamicin 2 mg/kg IV or IM (loading dose), followed by 1.5 mg/kg every 8 hours (maintenance dose); can be substituted with single daily dosing (3–5 mg/kg once a day)
	Regimen B: Cefotetan 2 g IV every 12 hours OR Cefoxitin 2 g IV every 6 hours PLUS Doxycycline 100 mg orally or IV every 12 hours	
Oral or IM‡	Regimen A: Ceftriaxone 500 mg IM in a single dose‡	Regimen D§: Levofloxacin 500 mg orally once a day for 14 days

Treatment	Recommended Regimens	Alternative Regimens
	PLUS <u>Doxycycline</u> 100 mg orally twice a day for 14 days WITH <u>Metronidazole</u> 500 mg orally twice a day for 14 days Regimen B: <u>Cefoxitin</u> 2 g IM in a single dose with <u>probenecid</u> 1 g orally administered concurrently in a single dose PLUS <u>Doxycycline</u> 100 mg orally twice a day for 14 days WITH <u>Metronidazole</u> 500 mg orally twice a day for 14 days Regimen C: Other parenteral 3rd-generation cephalosporin (eg, ceftizoxime, <u>cefotaxime</u>) PLUS <u>Doxycycline</u> 100 mg orally twice a day for 14 days WITH <u>Metronidazole</u> 500 mg orally twice a day for 14 days	OR <u>Moxifloxacin</u> 400 mg orally once a day for 14 days WITH <u>Metronidazole</u> 500 mg orally twice a day for 14 days Regimen E: <u>Azithromycin</u> 500 mg IV once a day in 1 or 2 doses, followed by 250 mg orally once a day for a total duration of 7 days WITH OR WITHOUT <u>Metronidazole</u> 500 mg orally twice a day for 12 to 14 days

* Recommendations are from the Centers for Disease Control and Prevention. [Workowski KA, Bachmann LH, Chan PA, et al](#): Sexually Transmitted Infections Treatment Guidelines, 2021. *MMWR Recomm Rep* 70(4):1-187, 2021 doi:10.15585/mmwr.rr7004a1

† Intramuscular or oral therapy can be considered for mild to moderate acute PID because the clinical outcomes with intramuscular/oral and parenteral therapy are similar. If patients do not respond to oral therapy within 72 hours, they should be reevaluated to confirm the diagnosis, and intravenous therapy should be given.

‡ If patient weighs > 150 kg with documented gonococcal infection, 1 g of [ceftriaxone](#) should be administered.

§ This regimen may be considered if the patient has a cephalosporin allergy, if community prevalence and individual risk of gonorrhea are low, and if follow-up is likely. Tests for gonorrhea must be done before therapy is started, and the following management is recommended:

- Positive culture for gonorrhea: Treatment based on results of antimicrobial susceptibility
- Identification of quinolone-resistant *Neisseria gonorrhoeae* or antimicrobial susceptibility that cannot be assessed: Consultation with an infectious disease specialist.

If patients do not improve after treatment that covers the usual pathogens, PID due to *M. genitalium* should be considered. Patients may be treated empirically with [moxifloxacin](#) 400 mg orally once a day for 7 to 14 days (eg, for 10 days).

Women with PID are usually hospitalized if any of the following are present:

- Uncertain diagnosis, with inability to exclude a disorder requiring surgical treatment (eg, appendicitis)
- Pregnancy
- Severe symptoms or high fever
- Tubo-ovarian abscess
- Inability to tolerate or follow outpatient therapy (eg, due to vomiting)
- Lack of response to outpatient (oral) treatment

In these cases, IV antibiotics (see table [Regimens for Treatment of Pelvic Inflammatory Disease](#)) are started as soon as cultures are obtained and are continued until patients have been afebrile for 24 hours.

Tubo-ovarian abscess may require more prolonged IV antibiotic treatment. Treatment with ultrasound- or CT-guided percutaneous or transvaginal drainage can be considered if response to antibiotics alone is incomplete (1). Laparoscopy or laparotomy is sometimes required for drainage. Suspicion of a ruptured tubo-ovarian abscess requires immediate laparotomy. In women of reproductive age, surgery should aim to preserve the pelvic organs (with the hope of preserving fertility).

Treatment reference

1. [Goje O, Markwei M, Kollikonda S, et al](#): Outcomes of minimally invasive management of tubo-ovarian abscess: A systematic review. *J Minim Invasive Gynecol* 28 (3):556–564, 2021. doi: 10.1016/j.jmig.2020.09.014

Key Points

- The sexually transmitted pathogens *Neisseria gonorrhoeae* and *Chlamydia trachomatis* are common causes of PID, but infection is often polymicrobial.
- PID can cause tubal scarring and adhesions, which commonly result in chronic pelvic pain, infertility, and increased risk of ectopic pregnancy.
- Because even minimally symptomatic infection may have severe sequelae, index of suspicion should be high.
- PCR and cultures are accurate tests; however, if results are not available at the point of care, empiric treatment is usually recommended.
- Hospitalize women with PID based on clinical criteria (see above).

Drug Information for the Topic



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